

Yara Elshehawi

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Education

B.Sc. in Media Engineering and Technology (Digital Media Engineering) Oct 2020 – Jul 2025

German University in Cairo (GUC)

Thesis: *Adapting a Human-Agent Toolkit to a Physical Robot Head (University of Augsburg, Germany)*

Experience

Machine Learning Intern — German University in Cairo Jul 2023 – Aug 2023

- Collaborated with researchers on data preparation for developing machine learning pipeline in a research lab environment
- Collected, annotated, and labeled 6,000+ images for weapons detection and age/gender classification tasks

Junior Teaching Assistant (Database I) — German University in Cairo Oct 2023 – Dec 2023

- Assisted students with SQL queries and relational database concepts during lab sessions

Key Projects

Lane Detection Using Semantic Segmentation 2026 *TensorFlow, Keras, OpenCV, Docker*

- Designed and trained a lane detection system using semantic segmentation (U-Net + ResNet-50) achieving 85% IoU
- Implemented image and video inference pipelines with lane visualization using TensorFlow/Keras.
- Applied data augmentation and model fine-tuning to improve generalization across road scenes.
- Deployed the model on Hugging Face Spaces using Docker with a web UI for video upload and visualization of full segmentation and lane-only outputs.

Medical Image Segmentation (Brain Tumor Detection) 2025 *Python, TensorFlow, OpenCV, Gradio*

- Trained a U-Net model on MRI scans to segment tumor regions
- Preprocessed medical images using OpenCV and evaluated using Dice score: 0.7019
- Deployed an interactive demo on Hugging Face Spaces using Gradio

Bachelor Thesis — Human-Robot Facial Expression Mapping 2024 *MediaPipe, Python, Omniverse, Flask*

- Developed a real-time facial expression tracking system using MediaPipe Blendshape models from webcam input
- Mapped facial blendshape coefficients to robotic actuators to achieve synchronized, human-like lip synchronization
- Integrated a Python-based pipeline with NVIDIA Omniverse for robotic simulation and visualization

Technical Skills

Programming Languages: Python, C++, Java, SQL, JavaScript, TypeScript

ML & Computer Vision: PyTorch, TensorFlow, OpenCV, MediaPipe

Web Development: REST APIs, Node.js, React

Cloud & Tools: AWS, Docker, Git, Hugging Face, Gradio

Achievements & Certifications

- DAAD Extra Scholarship Award, DAAD – Awarded to students with outstanding academic performance.
- Bachelor Thesis Scholarship, GUC – Awarded to the top-ranked student to complete my Bachelor thesis in Germany.

Leadership Experience

Model United Nations (MUN) *2023 – 2025*
Presidency of the International Court of Justice

- Progressed from Delegate to Board Member, serving two years in leadership roles
- Strengthened leadership, public speaking, and conflict resolution skills in high-pressure environments

Student Curriculum Committee — German University in Cairo 2023 – 2025

- Collaborated with academic staff to enhance overall student experience

Additional Information

Languages: Arabic (Native), English (Fluent), German (Basic)